



# THE GENESIS PROGRAM

Nurturing 21<sup>st</sup>-Century Leaders through

## PROJECT-BASED LEARNING

Grades 4–8

*NCP 2022 Aligned | NAIP 2025 Focused*



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## The Gap Every School Leader Already Knows

Can your students explain what they learned this week, in their own words, to someone outside the classroom? Most honest answers are uncomfortable, not because students or teachers aren't capable, but because the system rewards coverage over understanding.

NCP 2022 and Pakistan's National AI Policy 2025 both demand application, critical thinking, and problem-solving as explicit outcomes. Very few schools have a structured system to deliver them. That gap is what Genesis exists to close.

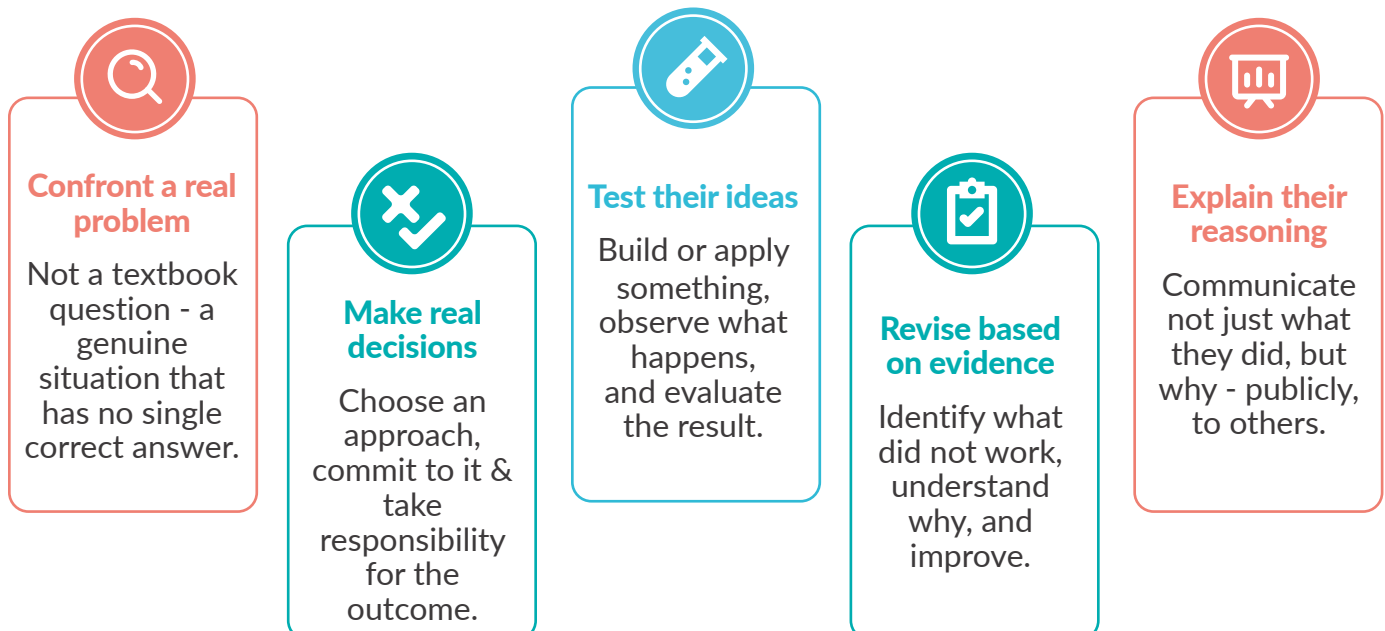
## The Solution: Structured Project-Based Learning

Project-Based Learning is not a new concept. Schools around the world have used it for decades. But it is often misunderstood and even more often, poorly implemented.

Many schools already run activities, exhibitions, and science fairs and these have real value. What Genesis builds is the deeper inquiry and decision-making layer behind them: a structure that ensures the project leads to genuine thinking, not just production.

### What real PBL actually is

Genuine PBL is not about what students make. It is about what students cannot avoid thinking through. A strong project creates a structured situation where students must:



When a project is designed properly, a student cannot simply copy an answer. They must think. And that thinking, repeated across projects, across grades, is what builds the capability that parents ask for and NCP 2022 demands.

## Who We Are

Genesis Education Solutions is a 21<sup>st</sup>-Century skills nurturing partner through structured Project-Based Learning built for Pakistan's schools, classrooms, and teachers, not a tutoring centre, a STEM vendor, or a workshop company.

We work inside your existing timetable and teachers, delivering your existing curriculum through a richer process.

Every project is built by educators for real Pakistani contexts, using locally available materials and NCP 2022/NAIP 2025 outcomes.

## The Genesis Program: What it is

A structured, year long Project-Based Learning implementation for Grades 4–8:

|                                                                                                                                                                                                                     |                                                                                                                                                                                                                             |                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p><b>6 projects per grade</b><br/>(5 NCP-aligned + 1 NAIP-focused), Schools implement any 4 across the year</p>                 |  <p><b>Multi-disciplinary</b><br/>3-4 subjects integrated per project</p>                                                                 |  <p><b>Classes</b><br/>10-12 Classes per project, maximum 5% of total academic sessions</p>           |
|  <p><b>Fully Resourced</b><br/>Student Booklets, Teacher Guide with rubrics, structured teacher training, year-round support</p> |  <p><b>No Special Infrastructure</b><br/>Existing infrastructure is sufficient - no extra labs, subscriptions, or imported equipment</p> |  <p><b>Certifications</b><br/>Students who meet the rubric standard earn the Genesis Certificate</p> |

### Why it works in Pakistani school classrooms

The common concern is that PBL sounds expensive, complex, or disruptive. Teachers worry about classroom control. Principals worry about curriculum coverage. School owners worry about parent perception. The Genesis approach addresses every one of these concerns directly.

## The Three Outcomes We Build



### Students become 21<sup>st</sup>-Century Leaders

Students move from waiting for instructions to acting as thinkers and solution builders, the 21<sup>st</sup>-century skills NCP 2022 and NAIP 2025 both demand.



### Teachers can teach through projects

Hands-on, project-specific training in facilitating structured, inquiry-driven learning, backed by a Teacher's Guide and rubrics for every project.



### Teachers build character, not coverage

Every project embeds decision-points where honesty, responsibility, and integrity are practiced, not just discussed.

## Why Schools Choose Genesis

- **Built for Pakistan:** National contexts, and NCP 2022/NAIP 2025 SLO mapped
- **Zero Disruption:** Runs inside the existing timetable
- **Teacher-Centered:** Makes teachers stronger, not replaceable
- **Character-Demonstrated:** Visible in the student's own work
- **Visible to Parents:** Every project ends with a real audience presentation
- **Results-Accountable:** Structured rubrics, collected booklets, certificates

Genesis Doesn't Deliver And Disappear. We Stay With The School Through Implementation.

## Team Behind The Genesis Program



Academic Lead  
**Muhammad Adil**  
 M.A. Economics & Finance

- 7 years at AKUEB - PBL curriculum design & implementation, teacher training and assessment frameworks
- 15 years faculty - Aga Khan Higher Secondary School, The City School, Beaconhouse, DM School System, Sadequain Grammar School



Academic Manager  
**Hasnain Murad**  
 MS Applied Mathematics

- 10 years faculty - The City School (PAF Chapter), Dawood Public School
- Fellowship in Holistic Science Teaching
- Microsoft Innovative Educator Expert
- Designs standard-aligned interdisciplinary STEAM learning



Senior Academic Associate  
**Muhammad Ibrahim**  
 PBL Alumnus

- PBL Alumnus, The Lab School (TLS)
- Organiser: Robotron 1.0 & 2.0
- Participant: Procom'24, Teknofest & Gadgeteer
- Judge: Robotron 3.0
- Currently pursuing Masters in Islamic Jurisprudence (2nd year)



Academic Associate  
**Muhammad Zain**  
 Curriculum & Educational Psychology

- Diploma in Curriculum Development & Educational Psychology
- Product Developer at The Learning Circle
- Masters in Islamic Jurisprudence



Creative Associate  
**Muhammad Jawad**  
 Certified In Brand Creation

- 5+ years design & production experience
- Creative head & senior editor at Services & FMCG Organisations
- Product Marketing Frameworks

# The Formula: A Project in Practice

Design, make, and launch Pakistan's next great drink.

## Group Constitution:

4 or 5 Students in a group

## Classes:

10-12 Classes per project

## Subjects:

Science, Mathematics, English, and Islamiyat

## Driving Questions:

Mix it right, measure it fair: How can we create a healthy drink everyone can trust?

## Your Mission

Food scientists design every drink you have ever tasted, the concentration, the ratios, the ingredients. Now it is your group's turn.

You will design a formula for a new drink using real Pakistani ingredients. You will mix it, test it, label it, and share what you made with an audience that has real questions. Make a group of 4-5 students. Every group makes something different. At the end, your class runs an awareness campaign about honest food labelling - and the audience includes people from outside your classroom.

## What You Will Walk Away With

- A real drink you designed and made from scratch, in a bottle, with your original label on it.
- The experience of standing in front of a real audience, answering real questions, and communicating science in plain words that anyone can understand.
- The ability to explain every single ingredient in your formula using Science and Mathematics.
- Your own data, percentages, observations, results, that tells the story of how you built it.

## Stage 1: Taste First, Think Later

Pakistan has a problem. Walk into any shop and pick up a juice carton. It will say “30% fruit juice” or “natural flavour” or “no added sugar.” But can you trust it? Food safety authorities in Pakistan regularly find that drink labels are inaccurate - percentages are overstated, artificial dyes are unlisted, and sugar content is hidden. Most people who buy these drinks have no idea how to check.

By the end of this project, you will. But first - before you read a single word about mixtures - you are going to taste three drinks and start figuring out what is actually going on.



## Table 1: Taste the Drink

Taste each one carefully, then answer:

|                                           | CUP A | CUP B | CUP C |
|-------------------------------------------|-------|-------|-------|
| What does it look like?                   |       |       |       |
| What does it taste like?                  |       |       |       |
| Can you see any particles?                |       |       |       |
| Does it look the same from top to bottom? |       |       |       |

### Now Think:

Cup A contains one substance. Cups B and C both contain more than one substance mixed together. Science has a word for any substance made of two or more things mixed without forming something new. That word is \_\_\_\_\_

Did you notice that Cup B looks different than Cup C. What do you think cup B is a uniform solution or Cup C.

Cup B      uniform   or   different      (circle either of these two options)  
 Cup C      uniform   or   different      (circle either of these two options)

### Quick question:

Which cups are mixtures? How do you know? Write one sentence.

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Why do you think you can see through the solution in Cup B but you cannot see through the solution in Cup C?

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## Stage 2: The Science You Need

You just tasted three drinks and noticed real differences - things that look different, settle differently, taste differently. Science has names for all of those differences. Before you design your drink, research the 8 key ideas below. You will keep using them throughout this project.

Use your textbook or any other resources to research these key ideas. Write your own understanding in the table below, in your own words.

Table 2: Key Terms

| Terms                 | Your explanation in your words | One example from today's taste test | Another example from daily life |
|-----------------------|--------------------------------|-------------------------------------|---------------------------------|
| Solution              |                                |                                     |                                 |
| Solute                |                                |                                     |                                 |
| Solvent               |                                |                                     |                                 |
| Homogeneous Mixture   |                                |                                     |                                 |
| Hetrogeneous Mixture  |                                |                                     |                                 |
| Suspension            |                                |                                     |                                 |
| Dilute solution       |                                |                                     |                                 |
| Concentrated Solution |                                |                                     |                                 |

Identify the drinks you tasted either as solution or suspension.

Cup A \_\_\_\_\_, Cup B \_\_\_\_\_, & Cup C \_\_\_\_\_

Which of the drinks you tasted are dilute, and which are concentrated?

Dilute \_\_\_\_\_

Concentrated \_\_\_\_\_

**Watch this video to check your understanding:**  
Mixtures and Solutions - What is the difference?

<sup>①</sup> <https://www.youtube.com/watch?v=i4ITvRNkRP4>

or

<sup>①</sup>



**Scan to  
watch the  
video**

Discuss and note down in your groups the key findings from the video.

## Stage 3: Design Your Formula

Every drink you have ever tasted was designed. Someone chose the ingredients, the ratios, the flavour profile. Now you do it.

Every great drink starts with what you have. Before you think of ideas, bring your ingredients to the table first. Then see what you can make with them.

*Footnote*

<sup>①</sup> This video is referred for educational purposes only. It is external and has been selected for its instructional value as part of the Genesis Education Solutions PBL program. Its content may not align with GES values viewers are advised to use their own judgment while watching. We take no responsibility for its content and all rights remain with the original content creator

## The Rules

**01** Discuss with your group before bringing anything.

**02** Each person brings a maximum of 2 ingredients.

**03** Your group may make more than one drink from the same set of ingredients.

**04** Measure all additions and subtractions precisely - no guessing

**05** Your drink must include at least one suspension element (something visible, like fruit pulp, seeds, or pieces).

**06** Use real Pakistani ingredients - fruits, spices, herbs, or natural sweeteners available at home or a local market.

*\* Your ingredients must be cost-effective & health-conscious*

## Table 3: Ingredients You Brought

List every ingredient your group has brought. Include who brought it, how much they brought, and whether it is a fruit/ juice or something else.

| Ingredient | Brought by | Amount / Quantity | Type (fruit/juice/spice/other) |
|------------|------------|-------------------|--------------------------------|
|            |            |                   |                                |
|            |            |                   |                                |
|            |            |                   |                                |
|            |            |                   |                                |
|            |            |                   |                                |
|            |            |                   |                                |
|            |            |                   |                                |
|            |            |                   |                                |
|            |            |                   |                                |
|            |            |                   |                                |

Research in your group the benefits of the ingredients used in the final drinks. Note your findings in the space below:

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### Table 4: Drink - Ideas From Your Ingredients

Look at your ingredients and brainstorm possible drinks. Try multiple combinations in your register with rough workings. Record here only the drink ideas that are actually workable with what you have

| Drink name/idea | Ingredients you will use from Table 3 | Practical? (Yes/Maybe) |
|-----------------|---------------------------------------|------------------------|
|                 |                                       |                        |
|                 |                                       |                        |
|                 |                                       |                        |
|                 |                                       |                        |
|                 |                                       |                        |
|                 |                                       |                        |

#### SAFETY



Wash hands and utensils before starting with the preparation of your drink. If using a heat source, your teacher supervises. Never taste anything unless your teacher confirms it is safe.

### Select and Make a Sample of Your Final Drink

Make different samples of your drink based on the ideas generated in Table 4: Try your drink ideas. Record any changes to ingredients during sample-making as rough working in your register. Come back to this kit only once you have tried and tested. Your final drink is the one that your group likes best and has the best but economical formula. Circle your selection in Table 4 above, then complete the working below.

☐ Your final drink name:

### Table 5: Plan Your Final Drink

Mention the details of your final drink in the table below:

| Ingredient | Why we chose it? | Solute or Solvent or Neither? | Will it dissolve? |
|------------|------------------|-------------------------------|-------------------|
|            |                  |                               |                   |
|            |                  |                               |                   |
|            |                  |                               |                   |
|            |                  |                               |                   |
|            |                  |                               |                   |



### Examine Your Drink

In Stage 1 you observed three cups and made sense of them. Now apply the same examination to your own drink.

**Table 6: Observe Your Drink**

| Observation                               | Your Drink |
|-------------------------------------------|------------|
| What does it look like?                   |            |
| What does it taste like?                  |            |
| Can you see any particles?                |            |
| Does it look the same from top to bottom? |            |

### What kind of mixture is your drink, and why?

Is it homogeneous or heterogeneous? Is it a solution, a suspension, or both? Look at your drink as a whole and explain based on what you can see. Be specific about each ingredient.

## Stage 4: Calculate & Plan

A good formula is precise. Food scientists measure everything. Now you calculate your ratios, work out your percentages, and design your label. The mathematics you are using here expresses one quantity as a percentage of another, simplifying ratios, writing continued ratios, and solving real-world word problems involving ratio and rate. These are the calculations your drink actually requires.

### Examine Your Drink

Your group will start by making 250 ml of your drink - enough for testing, tasting, and the pulp test. If your formula works well, you may scale up to a maximum of 1000 ml for the campaign. Decide your exact starting amounts for each ingredient and fill in the table.

**Before filling this in:** Pick up a juice carton or drink packet from home. Find the nutrition label on the back. Notice how each ingredient is listed as a percentage. Now ask yourself: does this look accurate? Could you verify it? That is exactly what you are about to do for your own drink - and the answer should always be: yes, anyone can check it.

Table 7: Final Formula

| Ingredient | Amount used                 | % of total drink                    | Soluble? (Yes/No) |
|------------|-----------------------------|-------------------------------------|-------------------|
|            | _____ ml/g                  | $\div \text{total} \times 100 = \%$ |                   |
|            | _____ ml/g                  | $\div \text{total} \times 100 = \%$ |                   |
|            | _____ ml/g                  | $\div \text{total} \times 100 = \%$ |                   |
|            | _____ ml/g                  | $\div \text{total} \times 100 = \%$ |                   |
| TOTAL      | _____ ml<br>(start: 250 ml) | 100%                                |                   |

Show one full percentage calculation below:

Example: 75 ml of pomegranate juice in 250 ml total  $\rightarrow 75 \div 250 \times 100 = 30\%$

My calculation:

### Now Work with Your Numbers

You have already calculated percentages in the table above. Now use those same numbers to go deeper - one idea at a time.

#### Part A - Ratio (comparing two ingredients)

A ratio compares two amounts. If you used 200 ml of juice and 100 ml of water, the ratio of juice to water is 200 : 100. You can simplify this by dividing both numbers by the same value:  $200 \div 100 = 2$ , and  $100 \div 100 = 1$ . So the simplified ratio is 2 : 1.

Now do the same with your two main ingredients:

My two main ingredients: \_\_\_\_\_ and \_\_\_\_\_

Amounts used: \_\_\_\_\_ ml and \_\_\_\_\_ ml

Ratio (not simplified): \_\_\_\_\_ : \_\_\_\_\_

Simplified ratio: \_\_\_\_\_ : \_\_\_\_\_

(Both divided by: \_\_\_\_\_)

**Part B - Continued ratio (three ingredients)**

A continued ratio compares three or more things at once. If your drink has juice : water : honey in amounts 150 ml : 100 ml : 50 ml, the continued ratio is 150 : 100 : 50. Divide all three by 50: you get 3 : 2 : 1.

Now do this for your drink's three main ingredients:

Ingredient A: \_\_\_\_\_ ml

Ingredient B: \_\_\_\_\_ ml

Ingredient C: \_\_\_\_\_ ml

Continued ratio (not simplified): \_\_\_\_\_ : \_\_\_\_\_ : \_\_\_\_\_

Simplified continued ratio: \_\_\_\_\_ : \_\_\_\_\_ : \_\_\_\_\_

(All divided by: \_\_\_\_\_)

**Part C - Proportion (scaling your formula)**

Proportion means keeping the same ratio but changing the total amount. If your drink is 300 ml and has juice: water in ratio 2: 1, then juice = 200 ml and water = 100 ml. If you want to make 600 ml (double), you multiply both by 2: juice = 400 ml, water = 200 ml. The ratio stays the same - the amounts grow.

Your class has \_\_\_\_\_ students. Each student gets 50 ml. Total you need: \_\_\_\_\_ ml.

My original formula total: \_\_\_\_\_ ml. Multiply everything by:

Scaled-up amounts:

Ingredient A: \_\_\_\_\_ → \_\_\_\_\_ ml

Ingredient B: \_\_\_\_\_ → \_\_\_\_\_ ml

Ingredient C: \_\_\_\_\_ → \_\_\_\_\_ ml

Did the taste of your drink change when you scaled it up? ☐ Yes ☐ No

Why or why not?

**Part D - Unit Conversion (ml and L)**

When you measure your drink in ml, you are already using the unit of capacity. But food labels and drink cartons also use litres (L). The conversion is simple: 1 L = 1000 ml. So to go from ml to L, divide by 1000. To go from L to ml, multiply by 1000.

Now use your own drink's numbers:

My starting batch (250 ml) in litres:  $250 \div 1000 =$  \_\_\_\_\_ L

My scaled-up batch (from Part C) in litres: \_\_\_\_\_ ml  $\div 1000 =$  \_\_\_\_\_ L

A standard 1.5 L bottle from a shop - how many ml is that? \_\_\_\_\_  $\times 1000 =$  \_\_\_\_\_ ml

Your label will show the total volume. Should it say ml or L? Decide which is clearer for the person reading your label, and write it here:

## Design Your Label

Every drink in a shop has a label. Most of them cannot be fully trusted. Yours will be different - every number on it comes from a calculation you did yourself. Your teacher will share a few examples of real Pakistani drink labels. Analyse them together: what do they tell you, and what do they hide? Then design your own label in the box below. It must include:

- ☐ Your drink's name
- ☐ Your team's name
- ☐ List of ingredients in order (most to least)
- ☐ The percentage of your ingredients
- ☐ A tagline (one sentence that makes someone want to try it)

Label Sketch, Draw your full label design here including all required elements

## Stage 5: Develop Your Product

You have a formula. You have a label. Now bring it to life. Make your drink - starting with your 250 ml batch - and scale up to at least 1000 ml once your formula is confirmed.

**Note:** You can reuse the water bottle for this purpose.

### SAFETY



Wash hands and utensils before starting with the preparation of your drink. If using a heat source, your teacher supervises. Never taste anything unless your teacher confirms it is safe.

### Step 1 - Taste your drink as it is now:

Describe the taste and texture in 1-2 words: \_\_\_\_\_

Discuss or research in your group on how the suspended material can be separated from the solution.

Write down your findings in the space below.

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### The Pulp Test

Once your drink has been tasted, now see that your drink has pulp or visible particles in it or not. Before you bottle your final drink, you are going to remove the pulp using filtration - and then find out what that does to the taste.

### Step 2 - Filter your drink:

Pour about 100 ml of your drink through a piece of filter paper or clean cloth into a second cup. Let all the liquid pass through. Look at what stayed behind.

**Table 8: Observe the Filtrate**

| What stayed in the filter? | What passed through (the filtrate)? | What does this tell you? |
|----------------------------|-------------------------------------|--------------------------|
|                            |                                     |                          |

### Step 3 - Taste the filtered drink.

Taste the filtered liquid. Compare the taste and texture to what you tasted in Step 1 (unfiltered drink).

Did the taste change? ☐ Yes ☐ No

Did the texture change? ☐ Yes ☐ No

Why do you think that happened?

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**Table 9: Summary**

Summarise the steps that you did during this stage and fill the table below:

| What you did | Observation while doing it | Any surprises? |
|--------------|----------------------------|----------------|
|              |                            |                |
|              |                            |                |
|              |                            |                |
|              |                            |                |
|              |                            |                |

## Stage 6: Launch Day

Your drink is ready. Your label is honest. Your data is real. Now you use all of it to do something that actually matters: run an awareness campaign about food labelling in Pakistan.

### Phase 1 - Research (Homework, before the next class)

You are about to speak in front of people who do not know what you know. Before you do, find out how to communicate clearly - by yourself, using any ideas and information gathered during the development of your drink.

### Phase 2 - Plan Your Campaign (In class, before Campaign Day)

Your teacher will show a short video of someone communicating clearly to an audience. Watch it together, then decide as a group how you will run your campaign. Your campaign format is your choice.

**Campaign format options (tick what your school approves):**

- ☐ Classroom presentation to an invited guest (parent, admin staff, another teacher)
- ☐ School assembly slot - speak to younger students or other classes
- ☐ Display table at school entrance or canteen - students stop, taste, and listen
- ☐ Community level speak to parents or community members at a school event

Our campaign format: \_\_\_\_\_

Our audience will be: \_\_\_\_\_

Now plan what each group member will say. Keep it simple - notes only, not a script:

**Table 10: Planning The Script**

| What we will say/show/demonstrate                                                       | Who will do it |
|-----------------------------------------------------------------------------------------|----------------|
| Explain the problem: what is wrong with many food labels in Pakistan?                   |                |
| Show our drink and label - let the audience see a label they can trust and verify       |                |
| Teach the audience one practical thing: how to check a label or spot a misleading claim |                |
| Invite questions from the audience and answer them                                      |                |

## The Moral Pulse

**Before your campaign, read this carefully.**

One day, the Prophet Muhammad ﷺ walked through a marketplace and passed a man selling grain. Prophet Muhammad ﷺ put his hand into the pile and found the grain underneath was wet, the seller had hidden damp grain beneath a dry surface to make it look better than it was.

The Prophet ﷺ said:

مَنْ غَشَّنَا فَلَيْسَ مِنَّا

"He who deceives is not of us."

Hadith 101 | Sahih Muslim

This was not just about grain. It was about the standard of Amanah, trustworthiness, that Islam requires in every transaction. Your label is a transaction. When someone reads your label, they are trusting that what is written is what is inside. The percentage you calculated, the ingredients you listed, that is your word.

**Think about this:**

- Looking at your final label is everything on it exactly what your test showed?  
\_\_\_\_\_
- If a company lists 'natural flavour' on a label but uses artificial dye, what has it done?  
\_\_\_\_\_
- The filtration test you ran in Stage 2 is the same kind of test food scientists use - separating what is hidden from what is visible. You now have the knowledge to ask: what is actually in this product?  
\_\_\_\_\_
- If your drink did not turn out as healthy as you hoped, would you still write the truth on the label? Why?  
\_\_\_\_\_
- Think of one product you or your family buys regularly - chips, juice, and biscuits. Do you think what is written on the label is exactly what is inside?  
\_\_\_\_\_

6. After doing this project, will you read food labels differently? What will you look for?

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### Phase 3: Campaign Day

Use your plan. Speak clearly. Make eye contact. Let the audience taste your drink and read your label. Then open it up - invite questions and answer them honestly. If you do not know the answer to something, say so. That is also part of the campaign: being truthful about what you know and what you don't.

After the campaign - record the audience's questions here:

**Table 11: Question & Answers**

| Question the audience asked | How we answered<br>(or: we said we didn't know) |
|-----------------------------|-------------------------------------------------|
|                             |                                                 |
|                             |                                                 |
|                             |                                                 |
|                             |                                                 |
|                             |                                                 |

## Stage 7: Reflect

Answer honestly. Not what sounds right, what is actually true for you.

1. In Step 1 you tasted three cups without knowing anything about mixtures. Now you can describe exactly what each cup was, chemically. Write the scientific description of Cup B and Cup C using all the vocabulary you now know.

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2. Look at your final formula (Table 7). Your drink has a percentage for each ingredient. If you were selling this drink and wanted to add 100 ml of extra water to your drink without changing anything else, what would happen to the concentration? What would change, and what would stay the same?

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3. Rafa-e-Aam means using your knowledge in service of others. Millions of Pakistani families buy labelled drinks every week without being able to verify what is inside. You now know how percentages work, what filtration can and cannot remove, and what a genuinely honest label looks like. Write one specific thing you could do - or tell someone - that would help a person in your family or neighbourhood make a more informed choice next time they buy a drink.

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## What I Made and What I Learned

Tick honestly the checklist below

| I can now...                                                                                                                | Not yet                  | Getting there            | Got it                   |
|-----------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| Describe what a mixture is and give 3 real examples.                                                                        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Explain the difference between homogeneous and heterogeneous mixtures.                                                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Define solute, solvent, solution, dilute and concentrated, & identify each in my own drink.                                 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Explain what filtration removes, and what it cannot remove.                                                                 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Calculate the percentage concentration of an ingredient using my own formula data.                                          | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Connect honest labelling to the Islamic concept of Amanah.                                                                  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Write a clear product label listing ingredients in order from most to least, with accurate percentages.                     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Measure ingredients accurately in ml and grams and record all quantities in a table.                                        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Explain to someone outside my classroom - using plain words - why food labels matter and what to look for when reading one. | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |



## Genesis Program Deliverables



### Teacher Training

Hands-on training, before the implementation



### Student Booklet

One per student, guides the thinking, becomes the evidence base for assessment



### Teacher Guide

Step-by-step facilitation for every participating teacher



### Assessment Rubrics

Evaluate students as they work, not just at the end



### Implementation Support

Genesis visits regularly throughout the year



### Evaluation & Certification

Booklets reviewed, Genesis Certificates issued to students who meet the standard

## How Assessment Works

"Genesis assessment combines qualitative and quantitative measures, evaluated across two parts.

**Qualitative assessment (30 rubric points)** is rubric-based, evaluated by the teacher: a portfolio review of the completed Student Booklet, a brief individual oral check, and a rubric evaluation across process, product, and 21<sup>st</sup>-century skills.

**Quantitative assessment (10 marks)** is based on multiple-choice questions covering content from every subject the project touched, provided by Genesis in the Teacher Guide for the teacher to administer.

The school's formal grading continues unchanged; Genesis adds a capability layer on top of it."

## The Implementation Roadmap

### 01 Context Review

Genesis reviews the school's structure, teacher capacity, and context.

### 02 Onboarding:

Genesis and the school finalise participation and implementation plan.

### 03 Teacher Training:

Structured & comprehensive training for participating teachers.

### 04 Project Selection:

School chooses its four projects per grade for the year.

### 05 Execution & Assessment:

Projects run inside the timetable; Genesis visits regularly.

### 06 Evaluation & Certification:

Booklets reviewed, Genesis Certificates awarded.

### 07 Review & Iteration:

Genesis and the school refine the next cycle together.

We are building this for the long term. Schools that embed the Genesis Program now will have a meaningful head start over those that wait. We would like your school to be one of them.

## THE GENESIS PROGRAM

*Transforming the quality of classroom learning,  
where learning meets Real-world practice.*



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